

The Elemental Reason

Light, Measurement, and the Minimal Conditions of Existence

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Abstract

This paper formulates the minimal conditions of empirical existence and argues that every physically distinguishable and empirically measurable reality endures only so long as three co-fundamental conditions remain operative: coherence, interaction, and complexity. The argument begins from a thought experiment on the suspension of the photon's motion and shows that the intuition of darkness is insufficient, because it still presupposes a physically operative world. Once the condition of energetic propagation itself is suspended, the analysis shifts to a more fundamental level, where it becomes clear that without interaction not only energetic relations lose their operative basis, but also the configuration of matter, the physical meaning of time, the order of spatial relations, and the status of mass and energy as operative magnitudes.

From this it follows that interaction is necessary but not sufficient. Real relation also requires preserved identity and distinguishable structure. Coherence, interaction, and complexity therefore emerge as the three minimal conditions of existence, condensed in the formula $E = C \times I \times K \neq 0$, where the zero of any term reduces existence to zero without possibility of compensation.

The paper then shows that the same structure appears with equal necessity in scientific practice itself. Every measurement requires an object that preserves sufficient identity, a real interaction with the apparatus, and an organized, interpretable result; the measuring instrument itself functions only within those same operative conditions. Scientific measurement therefore never captures a substrate stripped of the conditions of existence, but always real configurations of coherence, interaction, and complexity. Empirical reality thus appears as the continuous proof of work of this triad.

On this basis, The Elemental Reason emerges not merely as a formulation of the minimal conditions of existence and measurement, but as the Universal Law of Empirical Existence: the law that renders intelligible why every measurable reality endures only within C, I, and K, and why no scientific measurement has ever registered any reality outside them. At the same time, it establishes a methodological criterion of theoretical rigor: every entity, process, or structure claimed to be real must be articulable in terms of coherence, interaction, and complexity. In this form, The Elemental Reason is presented as both a Universal Law and the point of departure for a more fundamental reorientation of scientific inquiry.

1. Introduction — The Fundamental Question

The Elemental Reason formulates the Universal Law of Empirical Existence. It determines the minimal conditions without which no entity, no structure, and no process can endure as a physically distinguishable and empirically registrable reality. Empirical reality becomes measurable and articulable only so long as coherence, interaction, and complexity remain operative.

In its most condensed form, The Elemental Reason is expressed through the relation:

$$\mathbf{E \text{ (existence)} = C \text{ (coherence)} \times I \text{ (interaction)} \times K \text{ (complexity)} \neq 0}$$

This formula determines that existence depends upon the simultaneously operative presence of three conditions: coherence, interaction, and complexity. Coherence preserves the identity of an entity through time; interaction places it in real relation with the world; complexity sustains the minimal organization that gives it structure, distinguishability, and stable configuration. Together, the three constitute the minimal structure without which no distinguishable empirical existence can endure.

The point of departure of the argument is set in a thought experiment centered on the most elementary condition of physical interaction. The question that opens this inquiry is simple in formulation and fundamental in its consequences:

What happens if light stops moving?

This question shifts attention from the ordinary experience of illumination to the physical condition of energetic propagation. From it emerges the examination of interaction as a fundamental condition of reality. The mental annulment of this condition shows that reality cannot be thought as something given, but only as something dependent upon conditions that must remain operative. It is precisely at this point that coherence, interaction, and complexity assume the form of the three co-fundamental conditions of existence.

The same structure also emerges with the same necessity within scientific practice itself. Every measurement requires a system that preserves sufficient identity during observation, a real relation between the system and the instrument, and an organized and interpretable result. Every numerical value produced by measurement expresses a stabilized configuration of coherence, interaction, and complexity. The empirical object thus emerges as an operative configuration of C, I, and K.

From this point, the notion of *Proof of Work* takes shape. Every measurable reality appears as operative proof that C, I, and K are at work: an atom, a cell, an organism, a planet, or a galaxy endures as a real system only so long as these conditions remain

operative. The empirical universe thus appears as the general field within which they are realized in different configurations and at different scales of organization. From this there also follows a methodological consequence: every entity, process, or structure presented as real within scientific discourse must be articulable in terms of coherence, interaction, and complexity. The Elemental Reason thus emerges simultaneously as the Universal Law of Empirical Existence and as a methodological standard of theoretical rigor.

2. The First Experiment — The Physical Status of the Photon and the Insufficiency of Darkness

The question of what happens if light stops moving initially provokes the answer of darkness, drawn from ordinary experience, in which the absence of light is experienced as a lack of illumination. This response, however, presupposes a fully functional physical order: objects that can be illuminated, positions that can be distinguished, distances that can be traversed, and a world within which illumination may be present or absent as a relative state. Darkness marks an absence within a preserved order; it does not touch the condition that holds that physical order together.

The question posed here does not concern the relative absence of illumination, but the suspension of the photon's motion itself. The photon is not a quiet presence that, apart from itself, also possesses the capacity to move. It is physically constituted precisely in its propagation, in the sense that its physical identity is inseparable from the act of propagation. The moment this propagation is suspended, the notion of a stopped photon becomes physically empty: the term remains, but the content it claims to denote loses its basis, because there exists no coherent physical configuration corresponding to the expression "motionless photon."

It follows from this that darkness, as an optical category, does not grasp the problem at hand. It presupposes precisely what the experiment places in question: a world still physically operative, within which the condition of energetic propagation remains active even when actual illumination is absent. But once the photon's motion is suspended, the question is no longer which optical state would prevail. The question is whether the very condition of physical energetic connectedness — the condition without which neither illumination nor darkness retains physical status — can still remain operative.

3. The Annulment of Interaction — What Actually Collapses

3.1 The Suspension of the Photon as the Annulment of Energetic Transmission

The photon is not an object that carries energy from one point to another; it is itself the minimal form of electromagnetic energetic propagation. For this reason, the suspension of its motion does not affect a particular mechanism of transmission, but the condition without which electromagnetic energy ceases to be an operative magnitude. With the fall of this condition, every energetic connection between systems loses both its reality and its physical effectiveness.

3.2 The Loss of the Operative Basis of Physical Configurations

With the loss of real energetic connections between systems, there also falls away the operative basis upon which physical configurations stand as determinate and distinguishable wholes. Every physical configuration — from the atom and the molecule to the organism and the cosmic structure — stands only upon a network of real relations that sustains it as a determinate and distinguishable configuration. When interaction falls to zero, the operative basis of this network itself collapses, and with it collapses the status of physical configurations as real wholes. Atoms, molecules, and systems do not transform into another state; they lose the condition that allowed them to stand at all as distinguishable configurations.

3.3 The Insufficiency of the Category of Chaos

Chaos still presupposes distinguishable elements and a framework of reality within which disorder can be established as the state of a lost order. But the annulment of interaction reaches a more fundamental level: it removes the very condition that makes possible the distinction between order and disorder, between configuration and disintegration. Matter does not pass into a disorganized state within the same framework; it loses the condition without which no distinguishable state can be realized.

3.4 Time as a Dimension Dependent on Interaction

Time, in the physical sense, is articulated only through change. Change requires transition from one state to another, and this transition requires interaction, because only through real operative relations can one state produce or affect a subsequent state. When interaction falls to zero, no real transition remains, no articulable sequence remains, and no effective connection between states remains. In their absence, the distinction between “before” and “after” loses the very physical basis that had made it valid.

3.5 Space as an Order of Relations and Its Dependence on Interaction

Space acquires physical meaning only through distinction, position, and relation. For there to be a distinction between “here” and “there,” there must exist really distinguishable entities or states; for distance to exist, there must exist poles between which it can be determined; for position to exist, there must exist an order of distinguishable configurations in relation to one another. When interaction falls to zero, the operative basis of this order also collapses. With its collapse, space loses the condition without which the notions of distance, position, and extension no longer retain physical content.

3.6 Mass and Energy as Operative Magnitudes Dependent on Interaction

Energy has physical meaning only as an operative magnitude: only insofar as it is realized in propagation, transfer, transformation, and effect. Mass has physical meaning only insofar as it operates within an order in which inertia, gravitational influence, and configuration remain operative. When interaction falls to zero, energy loses its operative condition, because no process remains within which it can be realized; mass loses the field within which it can be distinguished and act as a physical magnitude. The annulment of interaction, therefore, leaves behind no hidden substrate, no unexpressed energy, and no mass frozen within an empty framework; it overturns the very framework within which mass and energy retain physical content. When interaction falls to zero, reality does not grow dark: it is annulled.

4. From Annulment to the Conditions of Existence

The annulment of interaction shows that existence cannot endure in its absence, thereby placing interaction in the category of necessary conditions. But the necessity of a condition does not imply its sufficiency: the fact that interaction cannot be absent does not yet show that its presence alone is enough to keep existence operative. For this reason, the question now assumes its full form: which conditions must be simultaneously operative for reality to continue?

4.1 Interaction as a Necessary but Not Sufficient Condition

Interaction presupposes the minimal identity of that which interacts. For real relations to exist between units, there must exist units that remain distinguishable during the very act of relation. A process in which nothing preserves sufficient identity to enter into and remain in relation is not interaction between distinguishable entities, but dispersion without an identifiable physical subject. The very physical meaning of interaction therefore requires, as its own condition, the presence of the identity of that which interacts.

4.2 Coherence as the Condition of Identity

Coherence preserves the identity of an entity through time, making it possible for it to remain distinguishable as a real unit throughout the processes, changes, and relations into which it enters. It does not require absolute immutability, but sufficient preservation of identity for the entity to remain what it is. An atom endures as an atom so long as it preserves its physical identity; a molecule endures as a molecule so long as it preserves the configuration that makes it distinguishable; an organism endures as an organism so long as it preserves the operative continuity of its biological identity. Without coherence, the very possible subject of interaction is lost, and with it the condition without which nothing can remain the same thing through time.

4.3 Complexity as the Condition of Structure

Coherence and interaction are not sufficient without the third condition: the minimal organization through which an entity can endure as a distinguishable configuration. Complexity, in its fundamental sense, does not imply a multiplicity of parts or a high degree of sophistication, but the presence of that minimal organization without which there is no configuration, no internal distinction, and no structure that can be preserved and enter into real relations. Even the most elementary entities presuppose this minimum of organization — the spin, charge, and mass of an electron, the electronic configuration of an atom, the architecture of the bonds of a molecule — while at higher levels the same condition appears in ever richer forms of organization. Without complexity, the very condition of structure is lost, and with it the basis of every distinguishable existence.

4.4 The Triad of Co-Fundamental Conditions and the Multiplicative Form of the Formula

Coherence, interaction, and complexity are co-fundamental conditions of existence, because each sustains a dimension without which the notion of existence loses its physical and ontological meaning: coherence sustains identity, interaction sustains real relation, complexity sustains configuration. None can replace the others. Coherence without interaction remains identity without physical effect; interaction without coherence remains relation without a distinguishable subject; both, without complexity, remain without structure. Therefore, the absence of any one of them overturns existence as a distinguishable reality.

This relation is expressed directly in the multiplicative form of the formula:

$$E = C \times I \times K \neq 0$$

The multiplicative form expresses what the additive form cannot express: the zero of any one term reduces the product to zero, without any possibility of compensation by the others. Zero coherence removes identity and, with it, every basis of existence; zero

interaction removes real relation and, with it, every physical efficacy; zero complexity removes configuration and, with it, every distinguishable structure. The formula condenses, in its most compact form, the minimal conditions without which existence cannot be sustained, either physically or ontologically.

5. The Structure of Measurement as the Manifestation of C, I, and K

The same threefold structure that emerges as a condition of existence also emerges, with the same necessity, within scientific practice itself. Scientific measurement does not take place outside coherence, interaction, and complexity, but only within them: the object of measurement must preserve sufficient identity to be tracked as an object, the instrument must enter into a real relation with it, and the result must assume an organized form in order to be registered and interpreted. The minimal structure of measurement therefore coincides with the minimal structure of real existence.

5.1 Coherence as the Condition of the Identity of the Empirical Object

The object of measurement must remain distinguishable throughout the entire interval of observation. A system that changes to such an extent that it can no longer be followed as the same system from one moment to the next ceases to be an empirical object. With this, the very basis upon which it can be determined what is being measured also collapses. This condition holds at every scale of scientific inquiry: a particle must preserve sufficient identity for its trajectory to be registered as the trajectory of a determinate entity; a chemical reaction requires that its substances and states be followed as parts of the same process; a biological organism must preserve sufficient continuity for the measured parameters to be related to the same system. Coherence is therefore a necessary condition not only in the ontological sense, but also in the methodological one: it keeps the object within the measurable field of science.

5.2 Interaction as the Condition of the Production of the Empirical Signal

No object becomes measurable without entering into a physical relation with the apparatus of observation. Only through this relation does the system produce a registrable effect: the telescope registers the photons that reach its detector; the thermometer enters into relation with the thermal energy of the body it measures; the particle detector produces a signal through its physical relation with the particles that strike it. Without interaction, no signal emerges, no value emerges, no result emerges. Interaction is therefore a necessary condition not only of every empirical procedure, but of its very possibility.

5.3 Complexity as the Condition of the Articulation of the Empirical Result

The physical effect produced by the interaction between system and instrument must assume an organized, registrable, and interpretable form. Without this form, there is no scientific result. The detector produces organized traces; the laboratory apparatus yields numerical series; the spectrometer yields a structured distribution of frequencies; the microscope yields a readable configuration of the observed object. In each case, the result acquires empirical status only through its organization. Complexity is therefore the methodological condition of articulation without which the physical effect becomes neither an interpreted result nor scientific knowledge.

5.4 The Measuring Apparatus as a Configuration of C, I, and K

Nor does the measuring apparatus itself stand outside the conditions that make measurement possible. It exists and functions only so long as it preserves sufficient structural coherence to remain a distinguishable instrument, enters into real interaction with the system it measures, and maintains an internal organization that makes it capable of producing, registering, and articulating a result. An instrument that does not preserve its own physical identity, that does not enter into real relation with the object, or that does not maintain sufficient organization to register and process the physical effect, is no longer a measuring instrument. The measuring apparatus thus emerges not as a neutral instance outside C, I, and K, but as a configuration of the same operative conditions that also make the empirical object itself possible. The measuring instrument therefore does not stand outside C, I, and K, but functions and articulates the result only within the same operative conditions that also make the object of measurement possible.

5.5 Numerical Value as the Expression of Configurations of C, I, and K

Every scientific measurement produces a numerical value. This value is not an arbitrary sign referring to a “matter in itself” stripped of the operative conditions of existence, but expresses a stabilized configuration of coherence, interaction, and complexity. A stone yields certain values of mass, density, and mechanical behavior because it endures as a determinate configuration of C, I, and K; another stone yields different values because it endures as a different configuration. The same holds for molecules, cells, stars, and every other measurable system. The variation of numerical values between systems does not express the variation of a hidden substance behind measurement, but the variation of the real configurations that measurement captures. The number produced by measurement thus expresses, in quantitative form, a real configuration of C, I, and K, and not a property of an object given in advance as an independent substrate.

5.6 The History of Science as the History of Measuring Configurations of C, I, and K

From this structure follows a general methodological conclusion: science has always measured configurations of coherence, interaction, and complexity. Measurement never unfolds outside them; the measured object never appears beyond them; and what is measured is, in every case, a stabilized configuration of this triad. C, I, and K thus emerge not only as necessary conditions of real existence, but also as the operative structures of every empirical practice. This dual status — ontological and methodological at once — establishes the bridge to the notion of Proof of Work: scientific measurement never captures a substrate stripped of the operative conditions of existence, but only real configurations of coherence, interaction, and complexity. Measurable empirical reality thus appears as the continuous operative proof that these conditions are at work.

5.7 From the Conditions of Measurement to the Conditions of Existence

The transition from the conditions of measurement to the conditions of existence does not rest upon an arbitrary identification of the epistemic plane with the ontological one. It rests upon a more fundamental fact: every claim about empirical reality requires that what is claimed to be real must be able, minimally, to stand as a possible source of effect, distinction, or articulation. An entity that cannot preserve sufficient identity, that cannot enter into any real relation, and that cannot sustain any distinguishable organization is not merely inaccessible to measurement; it is untenable as an empirical entity. For this reason, the fact that measurement requires coherence, interaction, and complexity expresses not only the limits of our cognitive apparatus, but the minimal conditions without which a system itself cannot endure as a physically distinguishable reality.

From this point, the limit of the Kantian objection also becomes intelligible. It may be said that C, I, and K are conditions of our access to reality, not conditions of reality itself. But the argument of this paper does not claim to penetrate to a “thing in itself” separated from every possibility of empirical articulation. Its claim is more limited and, precisely for that reason, more rigorous: there is no empirical reality outside the conditions that make possible its identification, interaction, and articulation. In this sense, coherence, interaction, and complexity are not presented merely as forms of the knowing subject, but as the minimal conditions under which a system can endure as a distinguishable reality within the field of empirical existence itself.

6. Proof of Work — The Universe as the Operative Proof of the Law

From the structure of measurement there follows a conclusion that is at once ontological and methodological: the empirical object does not present itself as a substrate given in advance and independent of the operative conditions of existence, but as a stabilized configuration that preserves identity, exercises interaction, and sustains organization. It retains its status as an empirical object only so long as coherence, interaction, and complexity remain operative within it.

This conclusion gives the notion of Proof of Work its rigorous meaning. The presence of an entity as a measurable reality constitutes evidence that the conditions making that reality possible are operative. The system does not emerge as empirical reality by virtue of a prior passive status, but because it preserves sufficient identity to endure as a distinguishable unit, enters into real relations that render it physically effective, and sustains organization that makes it configurable and measurable. Empirical existence therefore has an operative character: it does not endure as a frozen fact, but as a status maintained by the continuous operativity of C, I, and K.

This operative character appears at all scales of organization. The atom preserves its configuration because C, I, and K are operative at that scale; the organism preserves its own existence because the same triad remains operative within a higher order of complexity; the star preserves its structure and processes because those same conditions remain operative within another regime of organization; the galaxy preserves its form because the same structure is at work on the cosmic scale. The scale changes, the regime changes, the architecture of the configuration changes; the fundamental condition that sustains the configuration as empirical reality remains the same.

Accordingly, what measurement registers is not simply the presence of an object, but the presence of a stabilized configuration of C, I, and K. The object thus emerges as the empirical proof of their real operativity. If every empirical system constitutes a proof of work of this triad, then the universe presents itself as the widest field of its operativity: not as a multiplicity of objects placed upon a given substrate, but as an order of configurations in which C, I, and K remain operative across different scales and forms of organization. Every local system emerges as a manifestation of this general structure, and every empirical result makes it readable within a determinate configuration.

In this light, the history of science may be read as the history of measuring configurations of coherence, interaction, and complexity: every empirical advance has registered a form of their operativity, every successful theory has described the manner in which this triad remains operative in a determinate field of reality, and every numerical value produced by measurement has expressed a stabilized configuration of it. Proof of Work thus emerges

as a rigorous definition of the status of empirical reality: an entity is empirically real only so long as it endures as a stabilized configuration of coherence, interaction, and complexity, and its presence as a measurable object is the most direct evidence that these conditions are operative. Every empirical reality is a proof of work of C, I, and K.

7. Ontological Consequences — The Universe as Process

7.1 Existence as the Operative Continuity of Its Conditions

From the minimal conditions of existence and from the structure of measurement there follows a direct conclusion: existence is not static in character, but operative. It does not endure as the state of an entity given once and for all, but as a status maintained only so long as coherence, interaction, and complexity remain operative. Every physical system endures as a distinguishable system only so long as its identity remains preserved, its relations remain real, and its organization remains effective. With the fall to zero of any one of these conditions, the system loses its status as a distinguishable reality, because it loses the condition without which that status cannot be maintained. Existence therefore emerges as the operative continuity of the conditions that make it possible, and not as an inherent property of a substance given in advance.

7.2 Matter as a Stabilized Configuration of C, I, and K

The operative character of existence excludes any understanding of matter as a passive substrate. Matter does not present itself as a base upon which processes, relations, and forms are later imposed, but only as a stabilized configuration of coherence, interaction, and complexity. Only in this form does it emerge as a physically distinguishable and empirically measurable reality. The traditional notion of substrate presupposes a “matter in itself” that stands beyond the operative conditions of existence and precedes them. But what emerges in every empirical procedure is never this supposed substrate; it is always a determinate configuration of C, I, and K. Matter therefore acquires ontological clarity only when read as a stabilized form of these operative conditions, and not as an independent base from which they would be derived.

7.3 The Universe as an Order of Operative Configurations

From this shift there emerges the ontological meaning of the universe. The universe does not present itself as a frozen totality of objects that simply are, but only as an order of configurations in which coherence, interaction, and complexity remain operative. Every entity, every system, and every structure endures within this uninterrupted operativity. The universe, for this reason, does not endure as a completed state, but as a process: it maintains its own reality only through processes that preserve identity, propagate interaction, and stabilize organization across different scales and forms of architecture. The physical world thus emerges as the operative continuity of these conditions, realized

simultaneously from the level of elementary particles to cosmic structures.

7.4 Nothingness as the Absolute Absence of the Operativity of C, I, and K

Within this framework, the notion of nothingness acquires a precise meaning. Nothingness is not an alternative state of the world, nor the empty form of a preserved reality, nor an empty space within which nothing is found. All such representations still presuppose an operative framework of reality. Nothingness is the absolute absence of the operative conditions that make the world possible. So long as C, I, and K remain operative, existence endures as a distinguishable reality; with their fall to zero, what remains is not a world reduced to another state, but the collapse of the very condition through which any distinguishable state can be realized. Nothingness thus emerges not as an alternative to existence, but as the complete absence of the operativity of C, I, and K, that is, as its absolute limit.

7.5 The Status of the Leibnizian Question

Within this framework, the Leibnizian question — “Why is there something rather than nothing?” — does not receive an answer in the classical form of a causal explanation, but loses the very condition of its formulation. “Nothingness” does not emerge as an ontologically coherent alternative to existence, but as the absolute absence of the operative conditions that make any distinguishable reality possible. So long as nothingness is understood as the complete absence of the operativity of C, I, and K, it cannot endure as the opposite term of existence. In this sense, TER does not claim to solve the Leibnizian question in the traditional sense; it shows that the question itself dissolves once “nothingness” is examined with ontological rigor. It is not existence that collapses into nothingness; it is the very possibility that “nothingness” could stand as a coherent alternative to existence that collapses.

8. Methodological Implications — TER as Criterion and as a New Horizon of Inquiry

8.1 Numerical Value as the Expression of Operative Configurations

If every numerical value produced by measurement expresses a stabilized configuration of coherence, interaction, and complexity, then the empirical datum cannot be reduced to a local indicator of an isolated property. It emerges as a quantitatively readable expression of a real configuration and, as such, gives access to the concrete way in which C, I, and K remain operative in that system. When values change, the change has meaning not merely as the quantitative variation of an isolated magnitude, but as a sign of a reconfiguration of the operative relations that sustain the system as a system. This opens the possibility of a science of configurations, in which attention shifts from the classification of separate properties to the analysis of transitions between different

architectural regimes of C, I, and K. Phase transitions, biological organization, the emergence of new functions, and the formation of higher-level structures may thus be read as different forms of one and the same minimal interpretive horizon.

8.2 A Minimal Common Basis Across Disciplines

The operative structure of C, I, and K establishes a minimal interpretive level shared by empirical fields that operate through different methods and formalisms. A physical system, a biological organism, a cognitive network, and a technical structure are described in specialized terms and measured with different instruments; yet in each case what is measured are stabilized configurations of preserved identity, real relation, and distinguishable organization. This common basis does not abolish the differences between disciplines, nor does it replace their specific formalisms. It establishes a level at which empirical results may be compared and interpreted within the same ontological horizon. Physics, chemistry, biology, and neuroscience thus emerge as fields in which different configurations of the same minimal structure are registered, while the plurality of theories and methods is preserved upon a shared ontological reference.

8.3 The Reassessment of Contemporary Theoretical Programs

A considerable part of contemporary theoretical problems arises from the obscurity of the ontological point of departure. When reality is taken as a given substrate without examining the conditions that make it possible, when consciousness is treated as a problem detached from the general conditions of existence, or when laws are treated as the final level of explanation without analyzing the conditions that make systems possible, theoretical inquiry remains enclosed within the obscurity of its own initial point. TER does not call for the polemical negation of existing programs, but for a displacement of the point of departure. Every theory of consciousness, of the structure of physical reality, or of the architecture of laws must be evaluated by the way it articulates the minimal conditions of existence and measurement. The further it stands from this basis, the weaker becomes its connection to the ontological and empirical status of the claims it forms.

8.4 The Hard Problem of Consciousness

Chalmers's philosophical zombie is described as physically identical to the human being: with the same neural configurations, the same behaviors, and the same reactions. Yet it is precisely this description that overturns the experiment before it even begins. A system physically identical to the human being has the same configurations of coherence, interaction, and complexity. Its neural systems preserve the same coherence, enter into the same interactions, and sustain the same complexity, up to the level of self-modeling. From the structure of C, I, and K it follows that consciousness appears precisely at that degree of complexity at which the system models its own operative conditions. For this

reason, a system physically identical to the human being cannot lack consciousness. Chalmers's zombie is therefore not merely unobserved; it is ontologically impossible as an entity of measurable empirical reality. Its conceivability guarantees not its ontological possibility, but only its formulability in language.

This also reaches the very formulation of the hard problem of consciousness at its foundation. That problem stands only if one accepts as possible that two physically identical systems may diverge at the point of subjective experience. TER overturns this premise. Configurations of C, I, and K, once above the threshold of self-modeling, are sufficient for the emergence of consciousness. Experience does not enter as an external addition supervening upon physical configuration, but as a characteristic of that very level of organization. The question "why does this configuration produce experience at all?" presupposes that configuration and experience are separate and that their connection requires an additional explanation. But if that presumed separation is ontologically untenable, then the problem falls with it. The case of the zombie thus shows concretely that a thesis may be formulable in language and yet collapse ontologically before it can achieve the status of a sustainable theoretical alternative.

8.5 The Universal Law and the Criterion of Refutation

Four centuries of science have measured only configurations of coherence, interaction, and complexity. Every experiment has operated within them. Every numerical result produced by measurement has expressed their magnitudes. No empirical reality has ever been registered outside C, I, and K. It follows from this that The Elemental Reason formulates explicitly the universal law that scientific practice itself has exercised uninterruptedly in every one of its acts. Its status rests in the very common structure of all possible measurements. The only criterion for its refutation would be the discovery of an entity, process, or structure that exists as a distinguishable empirical reality outside coherence, interaction, and complexity, or that endures with one of them equal to zero. Absent such a counterexample, the universal fact remains untouched that every empirically measurable existence has emerged, emerges, and can emerge only within C, I, and K.

8.6 TER as a Methodological Criterion of Theoretical Rigor

TER may also be formulated as a methodological criterion of theoretical evaluation: every entity, process, or structure claimed to be real must be articulable in terms of coherence, interaction, and complexity. This criterion establishes a minimal standard of ontological and empirical intelligibility. A claim about reality has grounding only insofar as it can be situated within the operative conditions that make existence and measurement possible; without such articulation, its theoretical status remains untenable. This standard does not replace falsification, which sets the criterion of empirical testability, nor Ockham's Razor,

which sets the criterion of theoretical economy. It adds a more fundamental criterion ontologically: every claim must preserve a clear relation to the minimal conditions that make the world exist as a distinguishable and measurable reality. TER thus emerges simultaneously as a new horizon of inquiry and as a criterion of theoretical rigor: it formulates the minimal conditions of existence and establishes the basis upon which one may evaluate whether a scientific claim preserves a real connection to the conditions that make the world possible and measurable.

9. Conclusion — From a Question About Light to the Law of Existence

The question of the suspension of the photon's motion placed at the center of inquiry the most elementary condition of physical interaction. From it there immediately emerged the insufficiency of the intuition of darkness: darkness presupposes a world within which light remains a physical possibility, whereas the suspension of the photon's very motion shifts the problem to the condition through which energetic propagation remains operative — a level that precedes every optical state and every experience of illumination.

The collapse of this condition led the argument to its fundamental consequence. Without interaction, energetic transmission is interrupted; relations lose their operative basis; matter loses the condition of configuration; time loses the basis of change; space loses its physical meaning as an order of distinctions and relations; mass and energy lose the field within which they endure as operative magnitudes. From this chain it became clear that existence is not given in advance, but depends upon conditions that must remain operative.

From this conclusion emerged the minimal structure of existence. Interaction proved to be necessary, but not sufficient: real relation requires preserved identity and distinguishable structure. Coherence, interaction, and complexity thus assumed the form of three co-fundamental conditions, condensed in the formula:

$$E = C \times I \times K \neq 0$$

The multiplicative form expresses directly the necessary character of each condition: the zero of any one term reduces existence to zero, without any possibility of compensation by the others.

The same structure appeared within scientific practice itself. Every measurement requires an object that preserves sufficient identity during observation, a real relation between system and instrument, and an organized and interpretable result. Every numerical value produced by measurement expresses a stabilized configuration of coherence, interaction, and complexity. This places the triad not only at the foundation of existence, but also at

the foundation of the very possibility of empirical knowledge: science has no access to a reality outside C, I, and K, because the conditions of measurement and the conditions of existence coincide structurally.

From this point, the notion of Proof of Work took shape. Every measurable system appears as empirical evidence that coherence, interaction, and complexity are operative; the real object emerges as their stabilized configuration; and the universe presents itself as the widest field of the operativity of this triad. In this light, the history of science acquires a new reading: as the history of measuring configurations of C, I, and K across different scales and forms of organization.

From this there also follows the methodological consequence. If real existence and scientific measurement unfold entirely within C, I, and K, then this structure also acquires the status of a criterion of theoretical rigor. Every entity, process, or structure claimed to be real must be articulated in terms of coherence, interaction, and complexity; in the absence of such articulation, the connection of the claim to reality as existence and as a field of measurement remains untenable.

For this reason, The Elemental Reason emerges not merely as a formulation of the minimal conditions of existence and measurement, but as the Universal Law of Empirical Existence itself: the law that renders intelligible why every physically distinguishable and empirically measurable reality endures only within coherence, interaction, and complexity, and why no scientific measurement has ever registered any reality outside them. In this form, The Elemental Reason stands not simply as a conceptual framework or methodological proposal, but as a law grounded at once in the minimal conditions of existence and in the operative structure of scientific practice. It thus appears as both a Universal Law and the point of departure for a more fundamental reorientation of scientific inquiry.

Methodological Addendum

On the Analysis of Scientific Practice

The argument of this paper does not begin from a particular experiment, from a limited set of data, or from an isolated branch of science. It begins from the analysis of the operative structure that recurs in every empirical measurement. The reason the text relies so strongly on scientific practice is simple: physical reality is investigated only through measurement. In every empirical field, the system being measured must remain identifiable, must enter into real interaction with the apparatus or environment of observation, and the effect produced by this interaction must assume an organized form capable of being registered, compared, and interpreted.¹

It is precisely in this sense that the thought experiment concerning the suspension of the photon's motion must be understood. It does not replace empirical practice, but radicalizes it conceptually. By mentally suspending the minimal condition of the photon's propagation, the experiment no longer asks about a particular optical phenomenon, but about the condition without which measurement itself becomes impossible. At this point, the analysis of light and the analysis of measurement converge: both lead to the question of the minimal conditions that must remain operative for a system to endure as a physically distinguishable and empirically registrable reality.

These conditions are coherence, interaction, and complexity. Coherence preserves the identity of the system through time; interaction brings it into real relation through which it becomes measurable; complexity preserves the minimal organization without which there is neither a distinguishable configuration nor an articulable result. This holds not only for the object of measurement, but also for the measuring apparatus itself: the instrument too exists and functions as an instrument only so long as it preserves identity, enters into interaction, and maintains sufficient organization to produce a result. Scientific practice therefore encounters not a stripped substrate, but systems that must be identifiable, interactable, and articulable.²

From this there also follows the status of the numerical value. The number produced by measurement is not an arbitrary sign referring to a "matter in itself" stripped of the operative conditions of existence. It expresses, in quantitative form, a stabilized configuration of coherence, interaction, and complexity. The empirical datum thus emerges not as an isolated indicator of a detached property, but as a readable expression of the concrete manner in which C, I, and K remain operative in a determinate system.

¹Bas C. van Fraassen, *Scientific Representation: Paradoxes of Perspective* (Oxford: Oxford University Press, 2008).

²Ian Hacking, *Representing and Intervening: Introductory Topics in the Philosophy of Natural Science* (Cambridge: Cambridge University Press, 1983).

The philosophy of science has long analyzed parts of this structure. Van Fraassen has emphasized the representation of the identifiable structure of systems; Hacking has emphasized intervention and interaction as the basis of experimental practice; Galison has emphasized the stabilization of results through the relations between system, instrument, and interpretive framework; Cartwright has emphasized the dependence of scientific intelligibility upon organized contexts and not upon the self-disclosure of a stripped substrate.³ Yet these analyses usually remain on the epistemological plane: they clarify how science studies the world, but not whether the conditions it presupposes in practice are also the minimal conditions of empirical existence itself.

It is precisely here that the argument of this paper enters. It does not treat coherence, interaction, and complexity merely as methodological conditions of measurement, but as minimal operative conditions without which no physical system can endure as an empirically distinguishable entity. If measurement is possible only when these three conditions are simultaneously operative, then the very possibility of measurement reveals something fundamental about reality: that empirical existence is not given as a silent substrate, but is maintained as operative only so long as these conditions remain non-zero.

At this point, the notion of Proof of Work also becomes intelligible. If the object of measurement, the measuring apparatus, the signal, and the articulated result exist and function only so long as C, I, and K remain operative, then every act of measurement constitutes operative proof that these conditions are at work. Science never captures a reality stripped of coherence, interaction, and complexity; it always captures real configurations of them.

From this there also follows the final methodological consequence. TER serves not only to formulate the minimal conditions of existence and measurement, but also as a criterion for the evaluation of other theoretical claims. Every entity, process, or structure claimed to be real must be articulable in terms of coherence, interaction, and complexity. In the absence of such articulation, the connection of the claim to reality as existence and as a field of measurement remains untenable. In this form, TER emerges simultaneously as an explanation of the operative structure of science and as a minimal standard of theoretical rigor.

³Peter Galison, *How Experiments End* (Chicago: University of Chicago Press, 1987); Nancy Cartwright, *How the Laws of Physics Lie* (Oxford: Oxford University Press, 1983).

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